

TRAN THUAN HIEU

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EDUCATION

Hanoi Amsterdam High School for the Gifted, Hanoi

GPA: Grade 10: **9.3** Grade 11: **9.5** Grade 12 first semester: **9.9**

STANDARDIZED TESTS

SAT: **1570** | IELTS: **7.5** | AP Computer Science A: **5** | AP Calculus AB: **5**

AP Macroeconomics, Microeconomics, Physics 1, Statistics (2026)

ACHIEVEMENTS & HONORS

First Prize in National Round (rank 7/200), Third Prize in Asia Round, International Collegiate Programming Contest (ICPC)	2024
Champion (among 2.200 participants), Vietnam National Young Informatics Contest	2025
Champion (among 40 teams), Vietnam AI Championship	2025
Third Prize (top 8%), Vietnam National Olympiad in Informatics	2024
Second Prize (top 10/200), Vietnam Olympiad in Artificial Intelligence	2025
Gold Medal (top 5/300), Vietnam High School for Gifted Students Olympiad	2024
Top 3 Vietnam, Microsoft Imagine Cup Junior	2023
Silver Medal, World Invention Creativity Olympiad (WICO)	2024
Gold Medal & Talent Award, Central Highlands Informatics Olympiad	2024
First Prize in Scientific Product, Hanoi Young Informatics Contest	2023

LEADERSHIP EXPERIENCE

Founder & President, Moc Roi Water Puppet Business | [Ecommerce Store](#) | [Contract](#) 11/2023 - Present

A small business empowering deaf students through water puppetry

- Partnered with 5 artisans to train and support 25 deaf children producing 9,000+ handmade puppets; sold via Shopee e-commerce and outsourced contracts with tourism companies, generating \$36K revenue over 2 years.
- Allocated 50% profit to the deaf community and reinvested 50% to scale operations, with a long-term plan to transfer full ownership to Hanoi deaf education center.

Vice President & Head of Expertise, Illuminating | [Photos](#) Every summer from 2023 - Present

STEM & Coding teaching project for deaf students

- Led 30 volunteers to run weekly STEM classes for 50+ deaf students, building a structured learning roadmap in game design (Delightex) and LEGO Mindstorms coding, plus a Month of STEM with hands-on workshops.
- Supported students in creating 25+ imaginative 3D game projects and hosting weekly robotics challenges.

Director & Performer, Harmonies of the Crescent Lake | [Video Recap](#) 5/2025 - 8/2025

Art shows promoting water puppetry for deaf students

- Led a 20-member team to script and perform a 15-act puppet show; served as the main performer in 3 acts (25 minutes per show), delivering 3 live shows for 60+ teachers and students at the Hanoi Deaf Education Center.

Co-Admin and Co-Founder of HNOJ, Hanoi Online Judge | [Platform](#) 12/2022 - Present

- After hundreds of late-night Codeforces sessions due to timezone gaps, founded Hanoi Online Judge to offer better contest windows and localized problems, running 100+ contests and releasing 5,000+ problems for 2,000+ programmers over 3 years.

RESEARCH

Co-author, Sign Language Recognition: A Large-scale Multi-view Dataset and Comprehensive Evaluation | [Paper](#) From 12/2023. Paper published in 5/2025

- Collected and curated a large-scale sign language video dataset over 3 months at the Hanoi Deaf Education Center (84,000 videos, 30 signers, 1000 glosses); manually labeled data and conducted training and inference experiments across multiple models, achieving a 19.75% accuracy improvement over prior single-view datasets.
- Research accepted at the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2025, an A*-ranked conference (per ResearchGate) and ranked 6th in Computer Vision by Google Scholar Metrics.

Author & Researcher, Optimizing Paraphrase Detection: From LSTM to Multi-Architecture Ensembles | [Paper](#) 2024

- Iteratively improved paraphrase detection system through multiple modeling and training techniques, achieving final accuracy increase from 0.75 to 0.89.

Author & Researcher, A Comprehensive Survey of Square Root Decomposition | [Paper](#) 2024

- Analyzed and synthesized multiple square root decomposition techniques, evaluating their trade-offs and identifying optimal approaches across different problem constraints and use cases.

PRODUCTS

Co-Inventor, Lead Programmer & Developer, LittleHands | [Demo](#) | [Download](#) | [Source Code](#) 03/2024 - Present

This application focuses on supporting people with hearing impairments to communicate effectively through sign language recognition system

- Built a mobile-first sign language translation app, one of the few solutions enabling Vietnamese Sign Language sentence-level translation, using Video Transformer Networks trained on my WACV-published sign language dataset.
- Support bidirectional translation (sign ↔ text ↔ sign-language animation) only use smartphone, without specialized hardware.
- Developed the system with FastAPI, PostgreSQL, and C#, delivering a minimalist, deaf-friendly interface. Earned 800+ downloads.

Co-Inventor, Lead Programmer & Developer, SignHelper | [Source Code](#) 11/2023 - 4/2024

SignHelper was the primitive web-based version of LittleHands

- Built an open-source text-to-sign language animation system for Vietnamese, motivated by the language's phonetic nature and reading challenges for deaf users; supports speech-to-text and OCR inputs.
- Deployed and supported self-setup pilots for 5 broadcast units, benchmarking performance against human sign language interpreters.

INTERNSHIP

Algorithm Optimizer Intern, AimeOpt of Aimesoft | [Company's Website](#) 6/2024 - 8/2024

The first company to develop multimodal AI with 200+ installations worldwide

- Built real-time rail data pipeline integrating IoT, sensors, ML to predict demand, optimize timetables & reduce congestion via digital twin simulation.

INTEREST

Vietnamese Water Puppetry: Performance and wooden puppet crafting; apprentice artisan at Dao Thuc Traditional Craft Village, attending weekly training sessions to refine carving, puppet-making, and performance techniques under professional artisans.

Sports: badminton, soccer; a dedicated supporter of Real Madrid.

TECH STACK

Competitive Programming: C++ & Codeforces top 3% Master

AI/Machine Learning: PyTorch Lightning, KERAS, FastAPI, ONNX

Full-Stack Development: NextJS, C#, TypeScript, PostgreSQL

Tool: Linux, Git